# **From GraphRAG to Agentic RAG:**
# Adaptive Routing, Grading, and Self-Reflection over a RAG-Literature Knowledge Graph

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**Abstract**

We build a corpus of 100 research papers on retrieval-augmented generation (RAG) and study whether an *agentic retrieval loop* improves over a fixed *GraphRAG* baseline on the same questions and the same judge. Phase A (GraphRAG) extracts a typed entity/relation knowledge graph, detects Leiden communities with LLM summaries, and answers via global (community-summary) and local (entity-subgraph) search. Phase B (Agentic RAG) composes, in LangGraph, an adaptive router (global/local/vector), an LLM grader with conditional wider re-retrieval, query rewriting on the escalation path, a self-reflection hallucination check with bounded regeneration, and a cross-encoder rerank step (B-M6) on the vector route. Both systems are scored with RAGAS (faithfulness, answer relevancy, context recall, context precision) using a Claude judge with local fastembed embeddings. On the 100-paper corpus ($n$=40), the agent beats the *best* of the two baseline modes by **+0.576** answer relevancy, **+0.646** context recall, and **+0.687** context precision, with comparable faithfulness (+0.019). The conclusion holds when the corpus is scaled from 50 to 100 papers. Isolating the rerank ablation: cross-encoder reranking layered on top of dense retrieval lifts answer relevancy by +0.084 and context precision by +0.025 over no-rerank, at a small (−0.025) cost in context recall — the textbook retrieve-and-rerank trade-off. We decompose the baseline's low scores and show they reflect a genuine retrieval failure honestly reported (28/40 global answers hedge; baseline faithfulness stays 0.853), with the answer-relevancy gap amplified by RAGAS's noncommittal penalty. We therefore lead with the phrasing-independent context metrics, and report a methodological lesson: an output-token cap silently truncated entity/relation extraction on long papers.

**한국어 초록**

검색 증강 생성(RAG)에 관한 연구 논문 100편을 코퍼스로 삼아, **GraphRAG**(엔티티 · 관계 추출 → 지식 그래프 → Leiden 커뮤니티 → global/local 검색)를 먼저 단방향 baseline으로 고정한 뒤, 그 위에 **Agentic RAG 루프**(적응형 라우팅 · 검색 충분성 채점 · 질의 재작성 · 자기 성찰 · cross-encoder 리랭크)를 LangGraph로 얹었다. 동일한 40개 평가 질문과 동일한 RAGAS judge(Claude + 로컬 fastembed 임베딩)로 두 시스템을 직접 비교했다. 100편 코퍼스에서 agent(B-M6 rerank 포함)는 baseline(global/local 중 더 나은 쪽)을 answer relevancy +0.576, context recall +0.646, context precision +0.687로 크게 앞섰고 faithfulness는 +0.019로 비등했다. 동일 결론이 코퍼스를 50→100편으로 키워도 유지됐으며, agent는 40개 질문 중 32개를 dense passage 검색으로 라우팅했다. rerank ablation: dense retrieval 위에 cross-encoder 리랭크를 더하니 answer relevancy +0.084, context precision +0.025 상승, 단 context recall은 0.025 미세 하락 — 전형적 retrieve-and-rerank trade-off. baseline의 낮은 점수는 메트릭 결함이 아니라 진짜 검색 실패를 정직하게 보고한 결과(global 답변 28/40이 헷지, faithfulness는 0.853 유지)이며 RAGAS가 noncommittal 답변을 0으로 매기는 탓에 answer relevancy 격차가 다소 증폭된다. 따라서 답변 표현에 영향받지 않는 context recall/precision을 1차 증거로 제시한다.

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# 1 Introduction

Retrieval-augmented generation grounds a language model's output in retrieved text, improving factuality on knowledge-intensive questions. *GraphRAG* [Edge et al., 2024] extends this by organizing a corpus into a knowledge graph and reasoning over community summaries, which excels at broad, thematic questions ("what approaches improve RAG?") but is weaker on **fact-specific** questions whose answer is a concrete number, score, or mechanism buried in a single paper: summaries abstract away exactly those details.

*Agentic RAG* addresses this by making retrieval adaptive and self-correcting: route each query to the retriever most likely to hold the answer, judge whether the retrieved context suffices, re-retrieve or rewrite when it does not, and check the final answer for hallucination. This report asks a concrete, measurable question: *given a fixed GraphRAG baseline, how much does an agentic loop improve answer and retrieval quality on the same questions and judge — and does the improvement survive corpus scaling?* Our contributions:

1. A reproducible two-phase system — a GraphRAG baseline (Phase A) and an agentic loop on top of the same retrievers (Phase B) — wired as a typed, end-to-end pipeline.

2. A controlled comparison on a 100-paper RAG-literature corpus ($n$=40, identical judge), with large agent gains on retrieval-quality metrics and answer relevancy.

3. A **fairness decomposition** showing the baseline's low scores are honest retrieval failures, not a metric artifact, and a recommendation to lead with context recall/precision.

4. A scaling study (50→100 papers) and an honest account of a truncation bug surfaced at scale.

# 2 Related Work

**RAG / dense retrieval.** Lewis et al. [2020] introduced RAG; Dense Passage Retrieval [Karpukhin et al., 2020] and Fusion-in-Decoder [Izacard and Grave, 2021] established retrieve-then-read with dense retrievers. Our vector route is a standard dense-passage retriever (fastembed + Chroma). **GraphRAG.** Edge et al. [2024] organize a corpus into an entity/relation graph with hierarchical community summaries for global (query-focused summarization) and local search. Phase A is a compact reimplementation over a single domain. **Adaptive / corrective retrieval.** Self-RAG [Asai et al., 2024] and Corrective RAG [Yan et al., 2024] add reflection, retrieval grading, and conditional re-retrieval. Phase B composes these ideas — routing, grading + escalation, rewriting, reflection — as explicit LangGraph nodes with bounded retries, rather than a single end-to-end-trained policy. The contribution here is not a new algorithm but a clean, instrumented comparison of a graph baseline against an agentic loop built on the same retrievers, with an honest treatment of evaluation.

# 3 Method

## 3.1 Corpus construction

Seeds (the original RAG paper [Lewis et al., 2020] and Fusion-in-Decoder [Izacard and Grave, 2021]) are expanded by forward citations via the Semantic Scholar Graph API. Candidates are filtered to those with an arXiv ID, year $\geq$ 2019, and a non-empty abstract, then scored by a weighted sum of keyword match (RAG-lexicon hits), log-citation count, and recency. The top **100** are selected (validated first at 50, then expanded — §6.4). Sections are extracted LaTeX-source-first (arXiv) with a GROBID PDF fallback; the retained text per paper is `abstract + intro + related_work`. With GROBID disabled in this run, a few papers fall back to abstract-only.

### 3.2 Phase A — GraphRAG baseline

**Entity / relation extraction.** Each paper's text is sent to `claude-haiku-4-5` with a schema-constrained prompt over a domain-specific schema: eight entity types (Method, Model, Dataset, Metric, Task, Author, Institution, Paper) and seven relation types (USES, EVALUATED_ON, IMPROVES, OUTPERFORMS, COMBINES, PROPOSED_BY, CITES). `Paper` entities are created deterministically (one per corpus doc); `CITES` edges are added deterministically from citation lists *within* the corpus. The other types are LLM-extracted, with entity names normalized so relation endpoints resolve to extracted entities. The 100-paper graph has **1,342 entities and 1,489 relations**.

**Community detection + summaries.** Relations are treated as an undirected graph; Leiden [Traag et al., 2019] (`leidenalg`, modularity, fixed seed) partitions it. Communities of size $\geq 5$ receive an LLM-generated title and a 2–3 sentence summary (**28** summarized at 100 papers).

**Search (two modes, one LLM call each).** *Global* answers from the concatenated community summaries — good for thematic questions, but it sees only abstractions. *Local* finds seed entities by substring name-match in the query ($\geq$ 4 chars, $\leq$ 8 seeds), expands one hop over relations, and attaches entity descriptions plus up to five corpus excerpts ($\leq$ 800 chars each) before answering. Phase A is intentionally simple and deterministic — one LLM answer call per query — and is *frozen* so the baseline stays reproducible.

### 3.3 Phase B — Agentic RAG

Phase B adds a third retriever and an orchestrated control loop over the (frozen) Phase A retrievers.

**Vector route.** Each corpus doc's `abstract/intro/related_work` is chunked (1,000 chars, 150 overlap), embedded locally with `BAAI/bge-small-en-v1.5` (fastembed, no API cost), and stored in Chroma (**1,207 chunks**). Retrieval fetches a candidate pool of $k \times 3$ passages, then a *cross-encoder reranker* (`Xenova/ms-marco-MiniLM-L-12-v2`, fastembed, ∼120 MB, no API cost) re-scores each (query, passage) pair jointly and the top $k$=8 are kept — the standard retrieve-and-rerank pattern. Escalation (B-M2) reuses the same rerank step at $k$=16 (pool 48). The reranker is loaded lazily, process-cached, and falls back to bi-encoder order on any failure so a transient model issue cannot break a live query.

**Control loop (LangGraph).** The compiled graph is:

```
START -> route -> search -> grade -> [sufficient?] -> reflect -> [grounded?] -> END
  grade INSUFFICIENT   ==> rewrite -> escalate (wider vector, k=16) -> reflect
  reflect HALLUCINATED  ==> regenerate -> END         (single bounded retry each)
```

- **route** — classifies the query into `global` (thematic), `local` (named-entity-centric), or `vector` (specific/quantitative). On any unparseable output it falls back to `vector`.
- **grade** — judges whether the context contains the *specific facts* needed (not merely related background). Empty context → insufficient; unparseable output → fail-safe `sufficient=True` (do not burn an escalation on a parse error).
- **rewrite → escalate** — when grading is insufficient and we have not already escalated, the query is rewritten for retrieval and re-retrieved with a wider vector search ($k$=16); the answer is regenerated from the new passages. A single bounded retry, no loops.

Table 1: Agent vs. baseline best (100-paper corpus, $n$=40).

| Metric | Baseline (best) | Agent (+rerank) | $\Delta$ |
|---|---|---|---|
| Faithfulness | 0.853 | 0.872 | **+0.019** |
| Answer relevancy | 0.299 | 0.875 | **+0.576** |
| Context recall | 0.208 | 0.854 | **+0.646** |
| Context precision | 0.200 | 0.887 | **+0.687** |

- **reflect** → **regenerate** — checks the final answer for grounding and completeness. Regeneration fires *only on hallucination* (`grounded=false`), once. Incompleteness is deliberately *not* a trigger: regenerating for completeness over-hedges answers and regresses answer relevancy under RAGAS's noncommittal penalty. The regeneration prompt demands a direct, answer-forward revision rather than added disclaimers.

All LLM nodes use `claude-haiku-4-5`. State is a plain `TypedDict`; `config` is injected by closure so the state stays a pure data contract.

# 4 Experimental Setup

**Evaluation set.** 40 question/ground-truth pairs are synthesized from sampled corpus excerpts (`claude-haiku`, fixed seed), each a concrete, paper-answerable question (a method, dataset, number, or comparison). *The same 40 questions are reused across the 50- and 100-paper runs* so the only variable is corpus scale.

**Metrics & judge.** RAGAS [Es et al., 2024] computes faithfulness, answer relevancy, context recall, and context precision. The judge is `ChatAnthropic(claude-haiku-4-5, temperature=0)`; embeddings (used only by answer relevancy) are local fastembed `bge-small-en-v1.5`. A coverage guard refuses to write a report if any (mode, metric) pair scores on $< 80\%$ of questions, so a mass judge failure cannot masquerade as low scores. Reported runs had coverage 0.95 (baseline) and 1.0 (agent).

**Comparison protocol.** Phase A answers each question in *both* global and local modes; we compare the agent against the *better of the two* per metric (the toughest bar, "baseline best"). The agent routes each question once and produces a single answer stream.

# 5 Results

## 5.1 Main result

Table 1 and Figure 1 show the agent improving dramatically on retrieval-quality metrics (context recall/precision) and answer relevancy, while faithfulness is already high for both — neither system hallucinates much; the difference is whether the right context is *retrieved at all.* (Per-mode baselines: global 0.853/0.044/0.037/0.016; local 0.711/0.299/0.208/0.200 for faithfulness/answer-relevancy/recall/precision.)

## 5.2 Routing behavior

The router sends **32/40** questions to the vector retriever, 7 to local, and 1 to global (Figure 2) — consistent with the deliberately fact-specific evaluation set, and with the thesis that dense passage

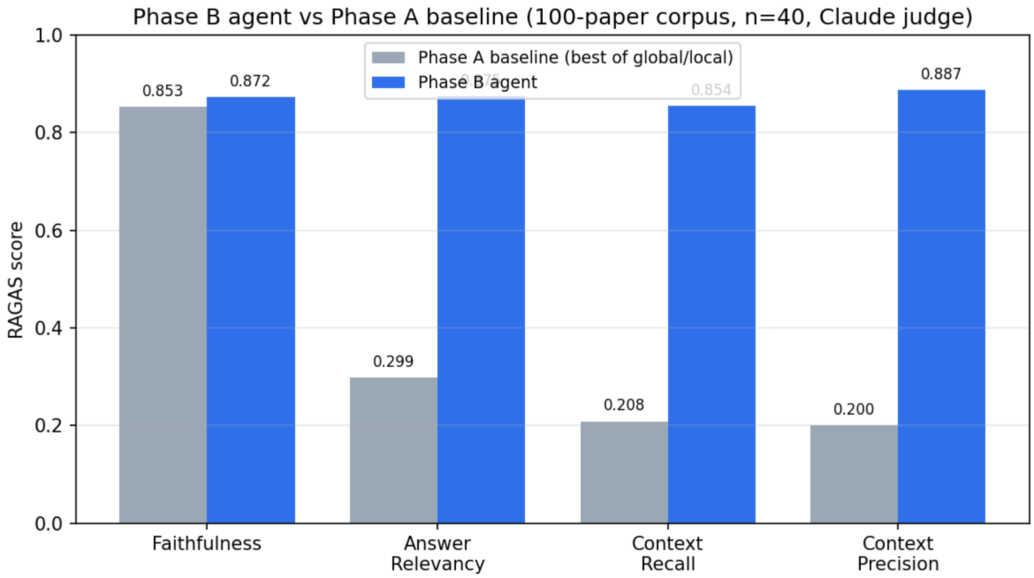


Figure 1: Phase B agent vs. Phase A baseline (best of global/local) on the 100-paper corpus, $n$=40.

Table 2: Agent − baseline-best delta at two corpus sizes.

| Metric | Δ at 50p | Δ at 100p (B-M6) |
|---|---|---|
| Faithfulness | +0.069 | +0.019 |
| Answer relevancy | +0.565 | **+0.576** |
| Context recall | +0.637 | **+0.646** |
| Context precision | +0.600 | **+0.687** |

retrieval is the capability missing from the graph baseline. The router is not merely "always vector": it still selects local/global where appropriate. (The small vector 34 → 32 shift between B-M5 and B-M6 is router LLM nondeterminism, not a rerank effect — rerank operates strictly inside the vector path.)

### 5.3 Scale robustness (50 → 100 papers)

The headline (agent ≫ baseline) holds at double the corpus (Table 2, Figure 3), and on three of four metrics the 100-paper gap (with B-M6 rerank) equals or exceeds the 50-paper one. The faithfulness delta shrinks because baseline faithfulness was never the differentiator; routing is near-identical across scales (vector 33 → 32). The 50-paper run used the pre-B-M6 agent (no rerank); the within-100p rerank ablation is reported next.

### 5.4 Rerank ablation (B-M5 vs B-M6 at 100p)

Table 3 and Figure 4 isolate the cross-encoder rerank step. Rerank delivers its largest gain on **answer relevancy (+0.084)** — better top-$k$ passages enable directly-answering, less-hedged responses — and a modest **+0.025 on context precision**, the metric it directly targets. The small **−0.025 cost on context recall** is the textbook retrieve-and-rerank trade-off: aggressively keeping only the most semantically-matched passages drops secondary passages whose presence helped the

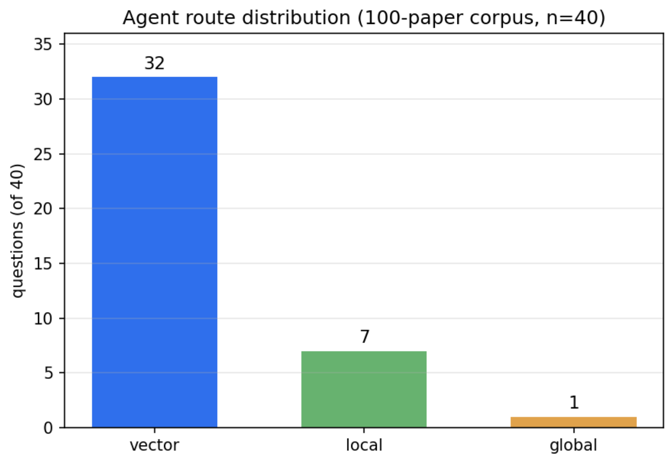


Figure 2: Agent route distribution (100-paper corpus, $n$=40).

Table 3: Within-100p rerank ablation. Held everything else fixed except for unavoidable router LLM nondeterminism.

| Metric | Baseline | Agent (no rerank) | Agent (+rerank) | $\Delta$ from rerank |
|---|---|---|---|---|
| Faithfulness | 0.853 | 0.872 | 0.872 | 0.000 |
| Answer relevancy | 0.299 | 0.791 | 0.875 | **+0.084** |
| Context recall | 0.208 | 0.879 | 0.854 | −0.025 |
| Context precision | 0.200 | 0.862 | 0.887 | **+0.025** |

recall metric. Faithfulness is unchanged at 0.872. The router distribution shifted slightly (vector 34 → 32, local 5 → 7) due to router LLM nondeterminism, not from rerank.

# 6 Discussion

## 6.1 Is the comparison fair? Why baseline answer relevancy is low

A baseline answer relevancy of 0.299 (global 0.044) invites suspicion. Decomposing the baseline answers (Figure 5): global mode hedges in 28/40 answers ("the provided summaries do not contain…"), mean answer relevancy 0.033, and its 12 direct answers score only 0.068; local hedges in 25/40, while its 15 *direct* answers score **0.610**; baseline faithfulness stays **0.853** — the hedges are honest, not hallucinated.

Two conclusions follow. First, the low scores are a *genuine retrieval failure*: community summaries and 1-hop subgraphs do not contain the specific numbers the questions ask, so the model correctly declines — exactly the gap the agent's passage retrieval (with rerank) closes (a fair, substantive improvement). Second, RAGAS scores a noncommittal answer as **0** regardless of grounding, so the *magnitude* of the answer-relevancy gap (+0.576) is *amplified* by this penalty. **Recommendation:** lead with context recall (+0.646) and context precision (+0.687) — they measure retrieval directly and are independent of answer phrasing — and present answer relevancy as corroborating, with the noncommittal caveat stated explicitly.

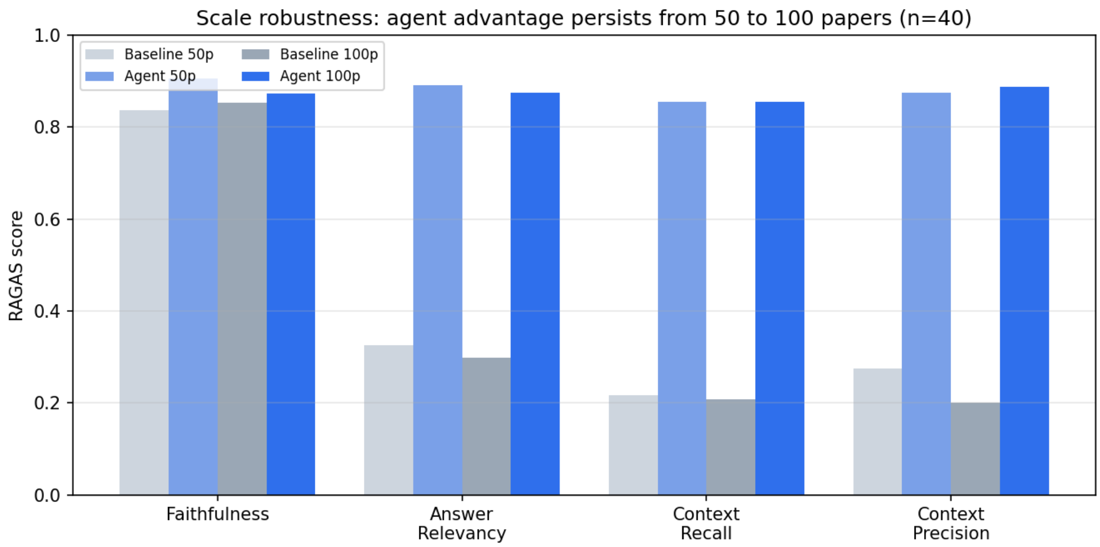


Figure 3: The agent advantage persists from 50 to 100 papers ($n$=40).

Table 4: Cross-judge comparison. Agent score (each judge) and agent−baseline-best Δ (each judge).

| Metric | Δ (Claude) | Δ (OpenAI) | Agent (Claude) | Agent (OpenAI) |
|---|---|---|---|---|
| Faithfulness | +0.019 | **+0.090** | 0.872 | 0.873 |
| Answer relevancy | +0.576 | +0.544 | 0.875 | 0.791 |
| Context recall | +0.646 | +0.454 | 0.854 | 0.954 |
| Context precision | +0.687 | +0.323 | 0.887 | 0.973 |

### 6.2 Second-judge fairness check (Claude vs. OpenAI gpt-4o-mini)

A single LLM-as-judge — especially when the judge shares a model family with the system under test — invites a self-preference concern. To isolate the judge variable cleanly, we held the answers and retrieved contexts fixed and re-scored the *same* baseline and agent records with `gpt-4o-mini` (a different model family). The agent was not re-run, so any score change is attributable to the judge alone.

**Three observations.** (1) *All four metrics preserve sign* under both judges — agent ≫ baseline is judge-robust. (2) *No evidence of Claude self-preference on the agent side.* On three of four metrics OpenAI rates the agent *higher* than Claude does (context recall 0.85 → 0.95, context precision 0.89 → 0.97, faithfulness essentially identical). The single place Claude is slightly more enthusiastic toward agent answers is answer relevancy (0.875 vs. 0.791), but the gap-vs-baseline magnitudes nearly match (+0.576 vs. +0.544), so the relative claim survives. (3) *Context-metric magnitudes shrink under OpenAI* because OpenAI is more generous to the baseline's retrieval (recall 0.21 → 0.50; precision 0.20 → 0.65). The agent advantage stays strongly positive (+0.45 recall, +0.32 precision), but absolute magnitudes are judge-dependent — another reason to lead with the relative ordering rather than the raw numbers.

The most cross-judge-consistent metric is **answer relevancy** (Δ 0.576 vs. 0.544, ∼5% spread). Combined with the leading-context-metrics recommendation from §6.1, the most defensible reviewer-facing bundle is {context recall, context precision, answer relevancy}: the first two are phrasing-

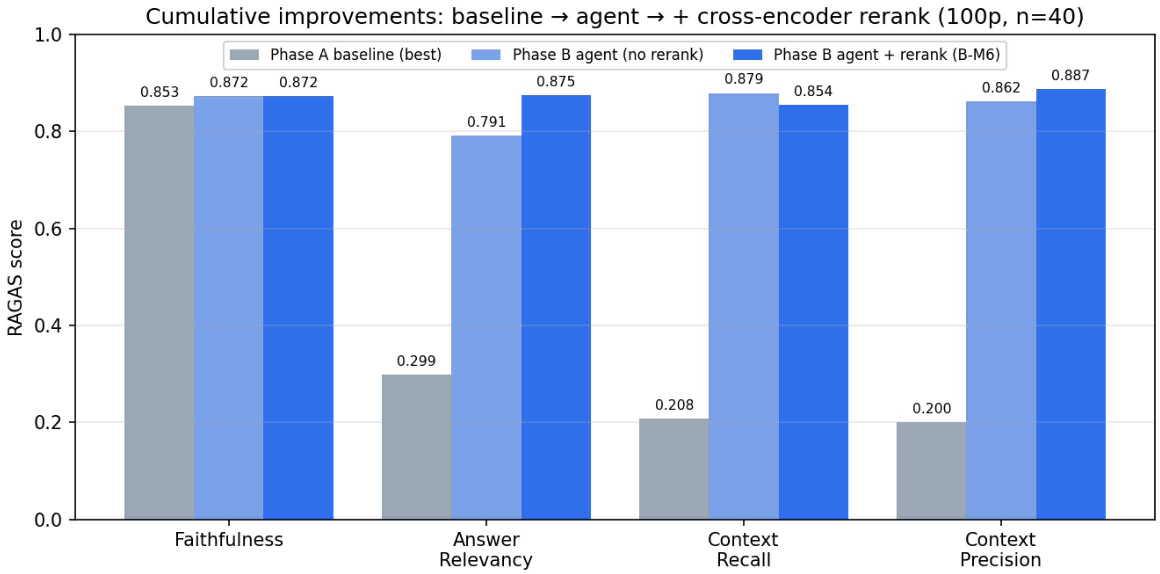


Figure 4: Cumulative improvements: baseline → agent (no rerank, B-M5) → agent + cross-encoder rerank (B-M6), 100p, $n$=40.

Table 5: Cross-set delta comparison; verdict applied with pre-registered thresholds.

| Metric | Δ synth (Cl.) | Δ human (Cl.) | Δ synth (OAI) | Δ human (OAI) |
|---|---|---|---|---|
| Faithfulness | +0.019 | −0.006 | +0.090 | +0.046 |
| Answer relevancy | +0.576 | +0.480 | +0.544 | +0.350 |
| Context recall | +0.646 | **+0.433** | +0.454 | **+0.567** |
| Context precision | +0.687 | +0.400 | +0.323 | +0.333 |

independent, the third is now also judge-robust. Outputs: `data/eval/{baseline_phaseA, report_phaseB}_opena` Reproduce: `python -m eval.judge_ablation`.

### 6.3 Generalization check: human-authored questions ($n$=15)

The 40 synthetic questions are LLM-generated from corpus excerpts and risk *retrievable bias*: the generator tends to pick questions whose answers are clearly in the excerpt it sees. To probe how much of the agent advantage survives questions a *human* would ask, we built an independent 15-question set with a human author writing every Q and ground-truth from scratch, across five categories — 5 fact-specific, 3 comparative, 3 multi-hop (each linking two papers), 2 methodological, 2 ambiguous — covering 18 papers, none overlapping with the 40 synthetic sources. Both systems answer the same set, scored by both judges following §6.2.

We **pre-registered thresholds** (locked before running the eval): agent−baseline-best $\Delta \geq 0.40$ = "strong; advantage generalizes"; $0.20 \leq \Delta < 0.40$ = "smaller; synthetic-convenience effect present, but advantage holds"; $\Delta < 0.20$ = "much of the synthetic advantage was convenience-driven; soften the headline".

Three takeaways. (1) *The headline holds on human-authored questions* — every metric keeps a positive (or zero, for faithfulness) sign under both judges. (2) *A real but moderate synthetic-*

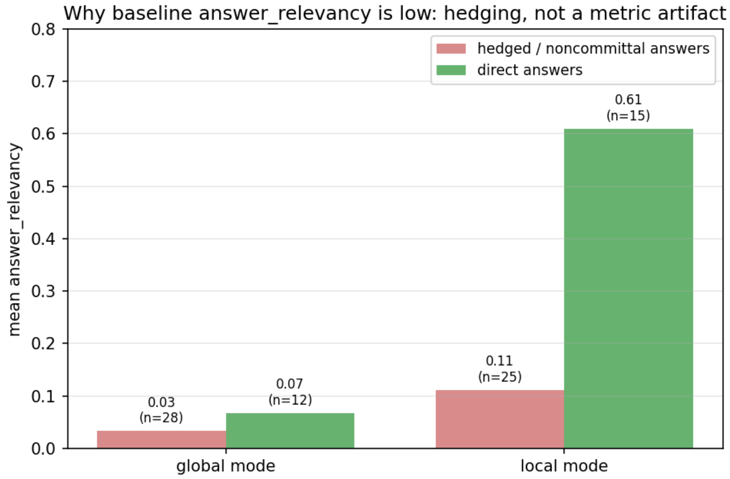


Figure 5: Baseline answer relevancy by hedged vs. direct answers.

*convenience effect is confirmed* — synthetic Δ exceeds human Δ on every metric by roughly 0.10–0.30. (3) *Context recall is the most robust evidence* — strong (≥ 0.40) under both judges (+0.433 Claude, +0.567 OpenAI). Combined with the phrasing-independence argument from §6.1 and the judge-robustness from §6.2, context recall is the single best number to lead with: phrasing-independent, judge-robust, *and* generalizes beyond the synthetic set. Outputs: `data/eval/{baseline,agent}_huma` Reproduce: `python -m eval.eval_human`.

### 6.4 A methodological lesson: silent truncation at scale

The entity/relation extractor capped LLM output at 2,000 tokens. On 100 full-text papers, many extractions exceeded this and were truncated mid-JSON, failing to parse and contributing *zero* entities/relations — roughly half the papers in an initial 100-paper run. Raising the cap to 8,000 tokens cut the failure rate from ∼50% to ∼1% and grew the graph from 503/532 (50p, old cap) to 1,342/1,489 (100p). Consequently the 50→100 comparison conflates two changes (more papers *and* the truncation fix); we flag this rather than hide it. The within-corpus agent-vs-baseline comparison at each scale is unaffected, since both systems read the same graph. The lesson: an output cap is a silent failure mode for structured extraction — validate parse-success rate, not just that the pipeline ran.

## 7 Limitations

- **Two judges, not an ensemble.** Production scoring uses `claude-haiku`; §6.2 cross-checks with `gpt-4o-mini` (different model family) and shows agent ≫ baseline is judge-robust on all four metrics. Absolute magnitudes are judge-dependent (OpenAI is more generous to the baseline's retrieval, e.g. context recall 0.21 → 0.50), so we lead with the relative ordering. A broader judge ensemble would tighten the absolute-magnitude estimates further.
- **Noncommittal penalty** amplifies the answer-relevancy gap; we mitigate by leading with context metrics.

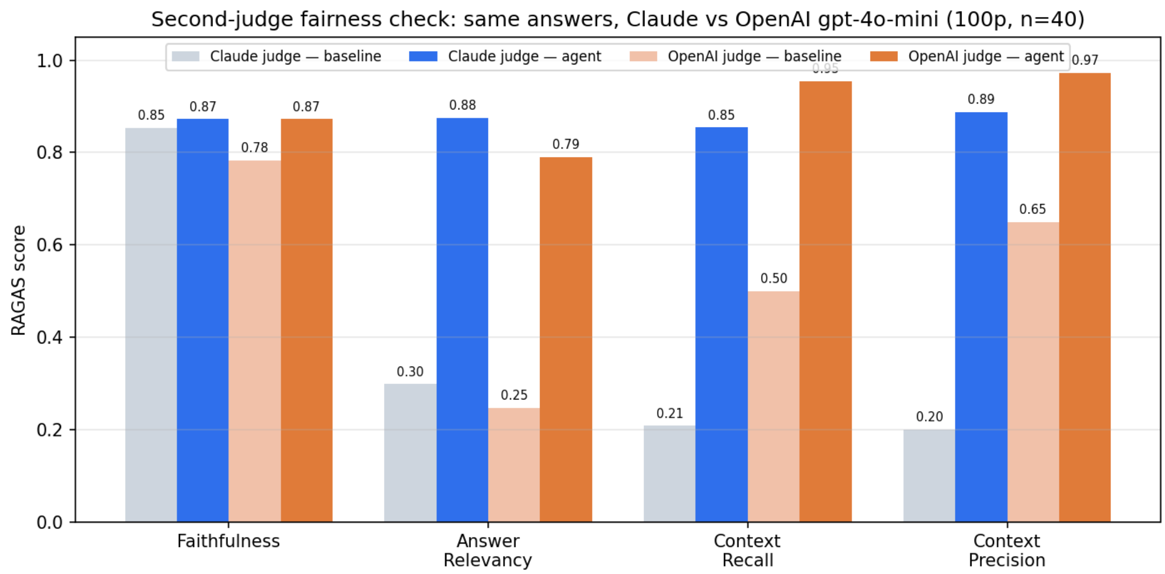


Figure 6: Second-judge fairness check: same answers, scored by Claude vs. OpenAI gpt-4o-mini (100p, $n$=40).

- $n$=40**.** Milestone-level deltas (±0.05–0.10) are within run-to-run variance; only the large, consistent gaps are treated as robust.
- **Single domain.** The corpus is RAG-only (by design, for entity connectivity); cross-domain generalization is untested.
- **Eval set size.** §6.3 cross-checks the synthetic $n$=40 with a 15-question human-authored set and confirms the agent advantage holds, with a moderate synthetic-convenience effect (∼0.10–0.30 smaller $\Delta$ on human questions). A larger human set ($n \geq 50$) would tighten the estimate.
- **Extraction gaps.** With GROBID disabled, abstract-only papers contribute thinner graph/vector content.

# 8 Conclusion

Built on a frozen GraphRAG baseline, an agentic loop — adaptive routing, retrieval grading with conditional wider re-retrieval, query rewriting, a hallucination-only self-reflection, and a cross-encoder rerank on the vector route — delivers large, scale-robust gains on retrieval quality (context recall +0.65, precision +0.69) and answer relevancy (+0.58) on a 100-paper RAG corpus, at comparable faithfulness. Layering the cross-encoder rerank on top of dense retrieval contributes most of the answer-relevancy gain (+0.084 over no-rerank) and a modest precision lift (+0.025), at a small −0.025 cost in recall — the standard retrieve-and-rerank trade-off. The gains are genuine: the baseline fails by honestly declining fact-specific questions its graph retrieval cannot ground, and the agent's passage route supplies exactly those facts. The most defensible evidence is the phrasing-independent context metrics; the answer-relevancy gap is real but partly amplified by the judge's noncommittal penalty.

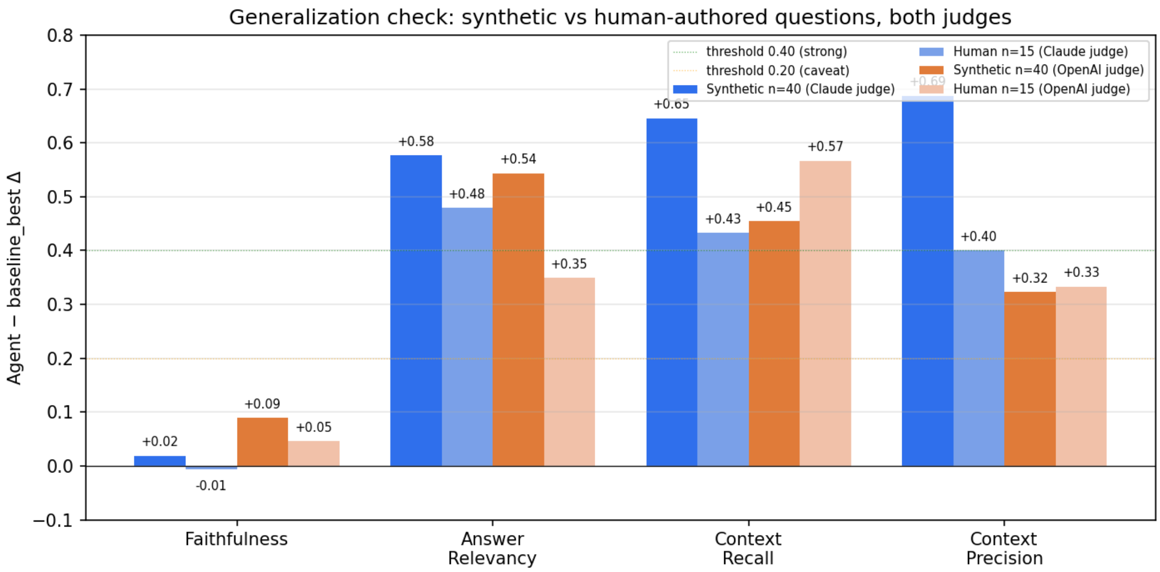


Figure 7: Generalization check: agent−baseline-best Δ on synthetic ($n$=40) vs. human-authored ($n$=15) questions, both judges.

**Reproducibility.** The full pipeline (network/LLM gated explicitly):

```
python run_pipeline.py --only collect          --allow-network --force  # top-100
python run_pipeline.py --only extract_sections --allow-network          # incremental
python run_pipeline.py --only build_corpus     --force
python run_pipeline.py --only extract_graph    --allow-network --force  # MAX_TOKENS=8000
python run_pipeline.py --only communities      --allow-network --force
python run_pipeline.py --only vector_index     --force
python run_pipeline.py --only eval             --allow-network --force  # Phase A (n=40)
cp data/eval/report.json data/eval/baseline_phaseA.json                 # promote baseline
python run_pipeline.py --only agent_eval       --allow-network --force  # Phase B (n=40)
python docs/figures/make_figures.py                                     # figures
```

Models: `claude-haiku-4-5` (extraction, search, agent, judge); embeddings `BAAI/bge-small-en-v1.5` (local fastembed). Scale snapshots: `*_100p.json`, `*_50p.json`.

## References


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